

Module 4: System Configuration

1. Introduction

This document provides the technical and functional specifications for the System Configuration module. This module serves as the central administrative hub for the entire ERP system, allowing authorized administrators to manage master data, user access, system-wide settings, and integrations. A well-structured configuration module is essential for maintaining data integrity, ensuring security, and adapting the system to evolving business requirements without requiring code changes.

2. Core Features

The System Configuration module will provide a secure, centralized interface for managing the following key areas:

- **User & Role Management:** A comprehensive system for creating and managing user accounts and defining their access rights and permissions across all modules.
- **Master Data Management:** A central repository for managing core business data that is shared across the system, such as the catalog of SEC certifications, the master list of materials, and standardized project data.
- **System Settings & Integrations:** A control panel for configuring system-wide parameters, such as notification triggers and external API endpoints.
- **Company Profile:** Management of the company's own information for use in reports and documents.

3. Detailed Tab Specifications

3.1. Tab 1: User & Role Management

This tab is the control center for system access and security.

3.1.1. User Interface

The tab will be split into two main parts:

1 Role Management:

- A grid displaying all existing roles (e.g., System Administrator, Project Manager, HR Manager, Warehouse Manager, Team Leader, Field Technician).
- Buttons to Add New Role and Edit Role.

- When adding or editing a role, a modal window will appear with a checklist of all system functions and permissions (e.g., Create Work Order, Edit Employee Profile, Process Material Receipt, Delete User). The administrator can select which permissions are granted to that role.

2 User Management:

- A grid displaying all system users with columns for Username, Full Name, Email, Assigned Role, and Status (Active/Inactive).
- A button to Add New User and Edit User.
- When adding or editing a user, a form will allow the administrator to set the user's details and assign them a role from the dropdown list of created roles.

3.1.2. Functional Logic

- The system will enforce Role-Based Access Control (RBAC) across all modules. Before any action is performed, the system will check if the current user's role has the necessary permission.
- Creating a new user will trigger an email to the user with a temporary password and a link to log in.
- Deactivating a user will immediately revoke their access without deleting their historical data.

3.2. Tab 2: Master Data Management

This tab allows administrators to manage the foundational data used throughout the ERP system.

3.2.1. Sub-Tab 2.1: HR Master Data

- **Course Catalog:** A grid to manage the master list of all SEC-mandated courses and certifications. The administrator can Add, Edit, or Deactivate courses. Each course entry will include:
 - Course Name (e.g., "Work At Height")
 - Validity Period (in months)
- **Course Requirements:** An interface to link courses to specific job roles or work types. For example, the administrator can specify that the "Excavation" work type requires the "Excavation Safety" and "Confined Space Entry" certifications.

3.2.2. Sub-Tab 2.2: Materials Master Data

- **Material Catalog:** A grid to manage the master list of all materials that can be collected from SEC. This data will pre-populate from the provided Warehouse.xlsx file but can be managed here. The administrator can Add, Edit, or Deactivate materials. Each entry will include:
 - SEC Material Number (e.g., 908111001)
 - Material Description
 - Unit of Measure (UOM)

3.2.3. Sub-Tab 2.3: Project Master Data

- **Districts:** A simple list to manage the names of districts where projects are located.
- **Municipalities:** A list to manage municipality names for permit tracking.
- **WBS & Material Templates:** (Optional Future Enhancement) A feature to create templates for common project types, pre-populating the WBS and Materials sections of a new Work Order.

3.3. Tab 3: System Settings & Integrations

This tab controls global system behavior.

3.3.1. User Interface

A form with various sections for different settings:

- **Notification Settings:**
 - Certification Expiry Alerts: Checkboxes and numeric inputs to enable and set the notification periods (e.g., Send alerts at 90, 60, and 30 days before expiry).
- **External System Integrations:**
 - A section for each external platform (Balady, Lab Portal, etc.) with fields for:
 - API URL
 - API Key / Credentials (stored securely, masked in the UI)

3.3.2. Functional Logic

- Settings saved here will be read by other modules to perform their functions. For example, the HR module's notification service will use the configured expiry alert settings.
- All credentials will be encrypted before being stored in the database.

3.4. Tab 4: Company Profile

This tab manages the company's own information.

3.4.1. User Interface

A simple form with fields for:

- Company Name (e.g., Barq/Alinjaz)
- Contractor ID (e.g., 1000329)
- Company Address
- Company Logo (Image Upload)

3.4.2. Functional Logic

- This information will be used to auto-populate fields in generated reports, certificates, and other official documents.

4. Database Schema

The following new tables are proposed for the System Configuration module:

Roles

Column Name	Data Type	Description
<u>RoleID</u>	INT (PK)	Primary Key
<u>RoleName</u>	VARCHAR(100)	Name of the role

Permissions

Column Name	Data Type	Description
<u>PermissionID</u>	INT (PK)	Primary Key
<u>PermissionKey</u>	VARCHAR(100)	Unique key for the permission (e.g., <u>work_order.create</u>)
<u>Description</u>	TEXT	Description of the permission

RolePermissions

Column Name	Data Type	Description
<u>RoleID</u>	INT (FK)	Foreign Key to <u>Roles</u>
<u>PermissionID</u>	INT (FK)	Foreign Key to <u>Permissions</u>

Users (This might extend the existing employees table or be a separate table)

Column Name	Data Type	Description
<u>UserID</u>	INT (PK)	Primary Key
<u>Username</u>	VARCHAR(100)	Login username
<u>PasswordHash</u>	VARCHAR(255)	Hashed password
<u>FullName</u>	VARCHAR(255)	User's full name
<u>Email</u>	VARCHAR(255)	User's email address
<u>RoleID</u>	INT (FK)	Foreign Key to <u>Roles</u>
<u>Status</u>	VARCHAR(50)	Active/Inactive

SystemSettings

Column Name	Data Type	Description
<u>SettingKey</u>	VARCHAR(100) (PK)	Unique key for the setting (e.g., <u>cert.expiry.alert.days</u>)
<u>SettingValue</u>	TEXT	The value of the setting

5. Integration Points

- **All Modules:** The System Configuration module is fundamentally integrated with every other module in the ERP system. User actions in all modules are first validated against the roles and permissions defined here.
- **HR Module:** Directly uses the Course Catalog and Course Requirements master data. The notification system for expiring certifications is configured here.
- **Warehouse Module:** Uses the Materials Master Data for validating and describing materials during receipt processing.
- **Work Order Module:** Uses the Districts and Municipalities master data in forms. It also relies on the external integration settings for platforms like Balady.